

Sai Sri Harsha Guddati

saisriha.guddati@ufl.edu — (352) 665-8709 — Gainesville, FL — <https://www.linkedin.com/in/sai-sri-harsha-guddati-552373180/> —
Portfolio: <https://saisriharsha.me> — Github: <https://github.com/saisriharsha19>

EDUCATION

University Of Florida

Master of Science in Computer Science, GPA: 3.9/4.0

Jan. 2024 – Present

Gainesville, FL

WORK EXPERIENCE

University of Florida, Information Technology

Artificial Intelligence and Platform Engineer Intern

Mar. 2025 – Present

Gainesville, Florida, US

- Built and deployed 15+ scalable AI assistants and chat tools across UF departments using OpenWebUI and Onyx, hosted on HiPerGator and UF's private cloud
- Fine-tuned Llama-70B, Gemma, and Mistral with domain-specific chat data and custom guardrails using Axolotl (LoRA / QLoRA + hyper-parameter sweeps), then served the models on-prem via vLLM—delivering 40% lower latency and 1.5% hallucinations across real-time apps
- Built custom web-crawlers to ingest domain-specific content and integrated them with DeepEvals and an LLM-as-judge loop, automating nightly evaluation of 10 K+ samples; the workflow accelerated testing 5× and improved response accuracy by 18% through iterative prompt-tuning and guardrail frameworks (SafeCompletion, Rebuff)

Tata Consultancy Services

Software Development Engineer

Aug. 2022 – Dec. 2023

Hyderabad, INDIA

- Led AI-powered projects leveraging Large Language Models (LLMs) integrated with Pinecone, Flask, and the OpenAI & open-source models for advanced use cases like programming language conversion, image analysis, and text recognition. Built scalable REST API's and a web interface using ReactJS and deployed it using Azure cloud, achieving a 40% improvement in up-time.
- Optimized database performance for PostgreSQL and MSSQL by implementing advanced indexing techniques, reducing query execution times up to 50%
- Developed solutions using Computer Vision models like YOLO for object detection and PyTesseract for OCR, processing 100,000+ images and text documents with robust testing frameworks like Pytest, ensuring 98% code coverage and seamless deployment.

PROJECTS

• VisionAI Suite: Scalable RAG-Driven LLM-Powered Data Processing and Object Recognition

Python 3.12 — Flask, OpenAI API, LangChain, Pinecone, RAG, Azure Cloud, YOLO, FastAPI, TensorFlow, PyTorch

- Built a multimodal retrieval app (LangChain + Pinecone, OpenAI NLP, Tesseract OCR) that boosted answer precision by 25%
- Created a high-performance FastAPI-based REST API with asynchronous execution and caching that improved uptime by 40%, paired with a React frontend enabling seamless real-time user interaction
- **AI Resume Ranker – Hybrid NLP + ML Resume Scoring Pipeline**
Python 3.8+ — Pandas, NumPy, scikit-learn, XGBoost, AdaBoost, Sentence-BERT, Flask, scikit-learn
 - Developed a hybrid resume ranking system leveraging transformer-based embeddings (SBERT) and ML ensembles (XGBoost, AdaBoost) trained on structured semantic and keyword-matching features
 - Engineered feature vectors using cosine similarity, TF-IDF scores, skills overlap, and experience heuristics to train interpretable ML models for candidate-job fit scoring and tuned models via cross-validation and grid search, achieving consistent precision in ranking top candidates; deployed system via Flask with real-time batch processing
- **GeoGuard: Privacy-Preserving Location Data Masking**
Python 3.12 — scikit-learn, Pandas, NumPy, GeoPandas, Folium, Matplotlib
 - Constructed a location masking application achieving 95% accuracy using KNN algorithms, while processing complex geospatial data with Pandas and NumPy for optimal transformation
- **Realtime Intelligent Patient Record System**
Python 3.12 — React — FastAPI, TypeScript, Hugging Face Transformers, Azure, CI/CD, Redis, MongoDB
 - Built a full-stack healthcare application using FastAPI (async), React/TypeScript and MongoDB to manage doctor–patient interactions and store longitudinal health records
 - Integrated LLMs (ChatGPT/Transformers) to summarize patient histories and doctor notes, storing outputs per user with Redis caching and PostgreSQL persistence for fast, context-aware follow-ups

SKILLS/ACHEIVEMENTS

Programming: Python (Advanced), Java, C++, Bash/Unix **Frameworks:** Flask, FastAPI, PyTorch, TensorFlow, React, Angular **Cloud:** Azure, Google Cloud, Azure Kubernetes, Azure ML Studio **Data Tools:** NumPy, Pandas, PostgreSQL, MSSQL, PL/SQL **Data Visualization:** Matplotlib, Seaborn, Tableau, Power BI **Dev Tools:** Docker, Git, REST APIs, PowerApps, Agile, Automation Scripts **Machine Learning:** RAG, LangChain, LlamaIndex, FAISS, Pinecone, Weaviate, pgvector, vLLM, Triton Inference Server, CNN, RNN, Transformers, LLMs, LoRA/QLoRA/PEFT, Axolotl, Prompt Engineering & Guardrails, Onyx agents, DeepEvals, Weights & Biases, Linear Models, Gradient Boosting (XGBoost/LightGBM), Deep Learning, NLP, OpenAI & Anthropic APIs, Hyperparameter Tuning, Docker, Kubernetes, FastAPI, MLflow **Web:** React, HTML, CSS, JavaScript, TypeScript